The reaction field, not the dielectric constant: solvent effects on the Menshutkin reaction $H_3N + CH_3Cl$

Extended version

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Note. This is the extended version of a report submitted for session 7 of the Computational Chemistry course, where a two-page limit applied. The argument, the data and the conclusions are the same; what the submitted version could not contain were the derivations, the secondary observables, the failed calculations and the discussion of the method's limits. All input and output files, scripts and figure sources are in the repository.

1 Introduction

1.1 The reaction

In a Menshutkin reaction an amine attacks an alkyl halide to give a quaternary ammonium salt [1]. The mechanism is a textbook S_N2: backside attack, inversion at carbon, a single concerted step. What makes it unusual is its charge balance. A conventional S_N2 such as $F^- + CH_3Cl \longrightarrow CH_3F + Cl^-$ begins and ends with one anion; charge is *displaced* from one atom to another and, at the transition state (TS), is *dispersed* over both. The Menshutkin reaction begins with two neutral molecules and ends with two ions:

$$H_3N + CH_3Cl \longrightarrow H_3N^{\delta+}\cdots CH_3\cdots Cl^{\delta-} \longrightarrow CH_3NH_3^+ + Cl^- \tag{1}$$

so charge is not displaced but *created* along the reaction coordinate.

The Hughes–Ingold rules [2] classify solvent effects on rate by exactly this criterion (Table 1). A polar solvent stabilises charge, and the more concentrated the charge the stronger the stabilisation; whichever of the two states, reactants or TS, carries the more concentrated charge is stabilised more, and the barrier moves accordingly. Creation of charge is the entry with the largest predicted effect, and experiment bears this out: Menshutkin rates span more than four orders of magnitude between hydrocarbons and dipolar aprotic solvents [3]. Historically this was the first systematic study of a solvent effect on a reaction rate, which is why Menshutkin's name is attached to a reaction class he did not discover but did quantify.

Table 1: The Hughes–Ingold rules. The two cases involving creation or destruction of charge produce large effects; redistribution of existing charge produces small ones. The reaction studied here is the first row; the $F^- + CH_3Cl$ reaction studied in session 4 is the third.

Change in charge from reactants to TS	Example	Polar solvent	Magnitude
Created	$H_3N + CH_3Cl$	accelerates	large
Destroyed	$R_3NH^+ + OH^-$	decelerates	large
Dispersed	$F^- + CH_3Cl$	decelerates	small
Concentrated	–	accelerates	small

1.2 The question the exercise asks, and the question it should ask

The exercise [4] asks for the activation barrier, the TS geometry and the charge on the leaving chloride to be computed in five solvents and correlated with the dielectric constant ε of the medium.

This is a reasonable request and it has a defect. ε is a macroscopic, phenomenological property of the bulk liquid: it is defined by the ratio of the capacitance of a condenser filled with the liquid to that of the same condenser in vacuum. It is not a quantity that appears, on its own, in any expression for the energy of a solute.

Where the solvent actually enters a continuum calculation is the reaction field. The solute is placed in a cavity carved out of a structureless dielectric; its charge distribution polarises the medium; the polarised medium produces a field that acts back on the solute; and the two are solved self-consistently [5]. For the two cases that are solvable analytically, the resulting stabilisation is

$$\text{ion in a sphere (Born):}\quad \Delta G_{\text{solv}} = -\frac{q^2}{2a}\left(1 - \frac{1}{\varepsilon}\right), \tag{2}$$

$$\text{dipole in a sphere (Onsager [6]):}\quad \Delta G_{\text{solv}} = -\frac{\mu^2}{a^3}\, f(\varepsilon), \qquad f(\varepsilon) = \frac{\varepsilon - 1}{2\varepsilon + 1}, \tag{3}$$

and the generalisation to a charge distribution of arbitrary multipole order in a spherical cavity, due to Kirkwood [7], has the same structure: each multipole order ℓ contributes a term proportional to $(\ell+1)(\varepsilon-1)/[\ell + (\ell+1)\varepsilon]$, of which Equations (2) and (3) are the cases $\ell = 0$ and $\ell = 1$.

Two features are common to all of them. First, ε never appears alone: it always appears inside a function of itself. Second, every one of those functions is *bounded*: the Born factor runs from 0 to 1, the Onsager factor from 0 to 1/2, and the general Kirkwood factor from 0 to $(\ell+1)/\ell$. The physical reason is that the reaction field is produced by the polarisation of the medium, and that polarisation saturates: beyond a certain ε the medium already behaves essentially as a perfect conductor, and making it nominally more polar adds nothing. The $2\varepsilon + 1$ in the denominator of Equation (3) comes from the boundary conditions satisfied by the potential at a spherical cavity surface.

The consequence for data analysis is immediate and is the subject of this work. ε is unbounded, running from 1 to about 110 for ordinary liquids; $f(\varepsilon)$ is confined to $[0, 1/2)$. The map $\varepsilon \to f(\varepsilon)$ is therefore severely compressive at the polar end (Figure 1). Acetonitrile and water differ by 43 units of ε and by 0.011 of $f(\varepsilon)$: to a continuum model they are almost the same solvent. Any quantity that is linear in the reaction field will therefore look like a saturating curve when plotted against ε, and that appearance is a property of the axis, not of the chemistry.

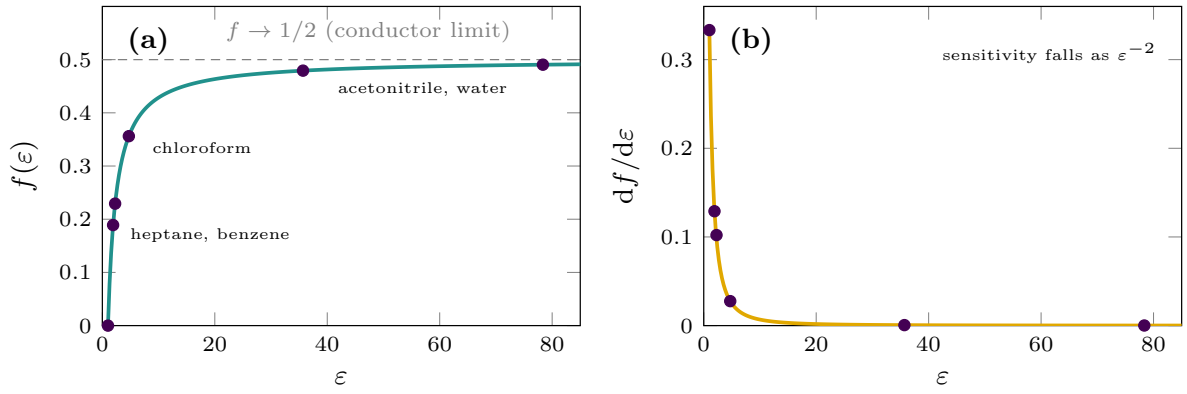


Figure 1: (a) The Kirkwood–Onsager function for the six media used here. The five solvents span only $0.19 \leq f \leq 0.49$ of the available range, and the two most polar are nearly coincident. (b) Its derivative: the sensitivity of the reaction field to ε falls as ε^{-2}, so beyond $\varepsilon \approx 20$ the dielectric constant has almost stopped carrying information.

1.3 Hypothesis

The hypothesis tested here is that the apparent saturation of solvent effects with ε is an artefact of the chosen variable. Specifically:

(i) Every quantity computed — barrier, reaction free energy, TS bond distances, charge on chlorine, dipole moment — should be a nonlinear, saturating function of ε and a *linear* function of $f(\varepsilon)$.

(ii) The resulting straight line should have predictive value: it should give the barrier in a solvent that was not used to construct it, from its dielectric constant alone.

(iii) Because a continuum model contains no solvent variable other than ε, the several correlations requested by the exercise should turn out to be a single correlation seen from different angles.

Equation (3) is used here to identify the variable, not to compute anything: all numbers come from the full polarizable continuum model with a molecule-shaped cavity, not from a dipole in a sphere.

2 Methods

2.1 Level of theory

All calculations were performed with Gaussian 09 [8] at the HF/6-31+G* level, the level prescribed by the exercise. The diffuse functions denoted by the + are not optional here: the product chloride is an anion whose density extends well beyond the valence region, and the TS carries a substantial charge separation. The polarisation functions on heavy atoms (*) are needed to describe the deformation of the carbon environment through the Walden inversion.

Hartree–Fock neglects electron correlation. This is a known and accepted limitation of the exercise, and its probable effect on the results is discussed in Section 3.8.

2.2 Choice of continuum model: PCM rather than SMD

Solvation was described with the polarizable continuum model [5], in the integral-equation formalism used by default in Gaussian 09. The cavity is built from interlocking atom-centred

spheres with radii scaled from van der Waals values; the solute polarises the surrounding dielectric; the polarisation is represented by apparent charges on the cavity surface; those charges generate the reaction field; and the cycle is iterated to self-consistency with the SCF.

SMD [9] was available and was deliberately not used. SMD computes the same bulk-electrostatic term and adds to it a second contribution — cavitation, dispersion and solvent structure — obtained as the solvent-accessible surface area multiplied by atomic surface tensions parametrised from solvent descriptors other than ε: refractive index, Abraham hydrogen-bond acidity and basicity, macroscopic surface tension, aromaticity and halogenicity.

For predicting an experimental solvation free energy, SMD is in general the more accurate of the two, precisely because those empirical terms absorb part of what specific solute–solvent interactions do. For the question asked here it is the wrong tool. The hypothesis under test is that a *single* electrostatic variable governs the solvent dependence; a model that contains several other solvent variables by construction cannot test it. Using the less accurate model is what makes the hypothesis falsifiable: with PCM, the only property of the solvent that enters the calculation is ε, so if the correlations fail, they fail for a reason.

2.3 Media

Five solvents were used — heptane, benzene, chloroform, acetonitrile and water — together with the gas phase as the $\varepsilon = 1$ limit. The gas phase is not decorative: it fixes $f(\varepsilon) = 0$ exactly, which anchors the origin of the abscissa. Without it the five solvents span only $0.19 \leq f \leq 0.49$, and a regression over that interval has a poorly determined intercept.

The dielectric constants used throughout are those applied internally by the program (1.9113, 2.2706, 4.7113, 35.688 and 78.3553 respectively), read from the output rather than taken from the handout, whose nominal values differ in the second decimal. Since the entire analysis is a regression against a function of ε, using values the calculation did not use would displace every point.

2.4 Species, chaining and convergence strategy

Six species were optimized and characterised by harmonic frequency calculations in each of the six media: the reactant complex $H_3N{\cdots}CH_3Cl$, the TS, the separated reactants NH_3 and CH_3Cl, and the products $CH_3NH_3^+$ and Cl^-. This amounts to 33 converged jobs. Every minimum gave only real frequencies and every TS exactly one imaginary frequency, whose normal mode is the S_{N2} itself.

The six media were chained in a single input file per species with `-Link1-`, each job reading the converged geometry of the previous one from the checkpoint file. The purpose is convergence, not convenience: a TS search is sensitive to its starting geometry, and stepping through the media in order of ε means each search begins from a structure optimized under a reaction field only slightly different from its own.

The direction of each chain was chosen by a simple principle: *start where the starting geometry is most reliable and end where convergence is least certain.* The reference geometries given in the handout correspond to a highly polar medium, so the TS series was run from water downwards to the gas phase, the gas-phase TS being the one most displaced from that reference. The reactant complex, by contrast, was expected to be most fragile in polar media, so its series was run upwards from the gas phase. Both expectations proved correct (Section 3.3 and Appendix A).

2.5 Reference state for the barrier

The exercise refers the barrier to the reactant complex. That reference turned out not to exist above $\varepsilon \approx 2.3$ (Section 3.3), so barriers here are referred to the separated reactants,

$$\Delta G^{\ddagger} = G(\mathrm{TS}) - G(\mathrm{NH_3}) - G(\mathrm{CH_3Cl}). \tag{4}$$

Besides being the only reference available in all six media, this is also the physically appropriate one for comparison with experiment, since Menshutkin kinetics are second order.

2.6 Atomic charges

Atomic charge is not a quantum-mechanical observable. There is no operator whose expectation value is "the charge on chlorine"; the electron density is continuous, and any division of it among atoms is a convention. Different conventions give different numbers, and the choice has to be justified.

The Mulliken scheme [10] partitions the density according to the basis functions it occupies: the population of each function is assigned to the atom on which that function is centred, and overlap populations between functions on different atoms are divided equally. Two things follow. The equal division is arbitrary and ignores electronegativity. More seriously for the present basis, the assignment depends on where a function is *centred*, not on where the density it carries actually is; diffuse functions extend far beyond their own centre, so with a + basis the assignment loses its meaning, and Mulliken populations are well known not to converge as the basis set is improved.

ChelpG [11] does not partition anything. It computes the molecular electrostatic potential from the converged wavefunction at several thousand grid points lying outside the van der Waals surface, and then finds by least squares the set of atom-centred point charges that best reproduces that potential subject to the total charge. The charges are defined by what they do rather than by how the density is divided. Their known weakness — that buried atoms are poorly determined, because their charge barely affects the external potential — does not apply to a chlorine atom at the periphery of the molecule.

Beyond numerical stability, ChelpG is the choice consistent with the problem. A continuum model generates its reaction field from the electrostatic potential of the solute; defining the charges by a fit to that same potential makes the quantity reported and the quantity driving the physics the same object. Both schemes are compared in Section 3.4.

2.7 Rate constants and data extraction

Relative rate constants follow from transition state theory. For two media at the same temperature the pre-exponential factors cancel, so

$$\frac{k(\text{solvent 2})}{k(\text{solvent 1})} = \exp\!\left(-\frac{\Delta\Delta G^{\ddagger}}{RT}\right), \tag{5}$$

with $RT = 0.592\,\mathrm{kcal/mol}$ at $298.15\,\mathrm{K}$. The same relation read backwards converts published relative rates into barrier differences, which is how the comparison with experiment in Section 3.7 is made.

Energies, free energies, optimized internal coordinates, ChelpG and Mulliken charges and dipole moments were extracted from the 33 output files with an `awk` script reproduced in Appendix B, and analysed with a Python script. Both are in the repository.

3 Results and discussion

3.1 The effect does not saturate; the axis does

Table 2 collects the results. Correlated against ε, as the exercise requests, not one of the six quantities is linear: the coefficients of determination lie between 0.45 and 0.58. Correlated against $f(\varepsilon)$, every one of them is a straight line, with R^2 between 0.977 and 0.9998. The points are the same, the calculations are the same, and only the abscissa changes (Figure 2a,b).

Table 2: Transition state geometry, ChelpG charge on chlorine, dipole moment, imaginary frequency, activation barrier (from the separated reactants) and reaction free energy in the six media. The last two rows give the coefficient of determination of a linear fit of each column against ε and against $f(\varepsilon)$.

Medium	ε	$f(\varepsilon)$	d(N–C) Å	d(C–Cl) Å	q(Cl) e	μ D	$\nu^{\ddagger}$ cm^{-1}	$\Delta G^{\ddagger}$ kcal/mol	ΔG_{r} kcal/mol
Gas	1.0000	0.000	1.899	2.474	−0.790	13.51	−559.1	47.67	114.09
Heptane	1.9113	0.189	2.035	2.396	−0.774	12.89	−574.0	37.78	51.43
Benzene	2.2706	0.229	2.061	2.381	−0.771	12.77	−572.2	36.11	40.59
Chloroform	4.7113	0.356	2.136	2.335	−0.761	12.37	−563.8	31.68	10.85
Acetonitrile	35.6880	0.479	2.203	2.292	−0.749	12.00	−554.5	28.24	−12.97
Water	78.3553	0.490	2.209	2.289	−0.748	11.96	−553.6	27.97	−14.93
R^2 vs. ε	–	–	0.510	0.527	0.576	0.549	–	0.447	0.464
R^2 vs. $f(\varepsilon)$	–	–	0.995	0.998	0.999	1.000	–	0.977	0.983

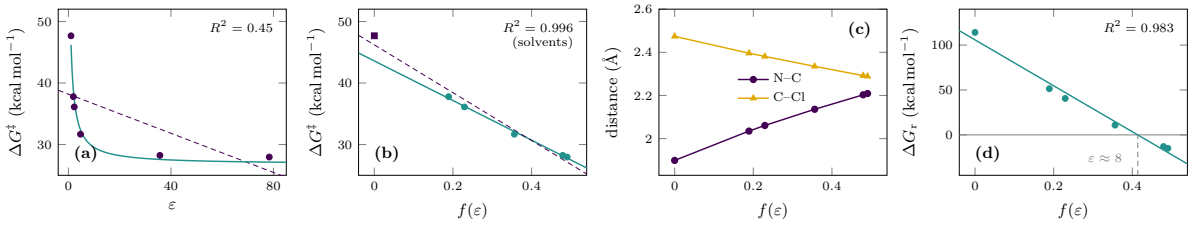


Figure 2: (a) Activation barrier against ε. The dashed line is the linear fit the exercise invites ($R^2 = 0.45$); the solid line is the model of panel (b) mapped back onto this axis, showing that the data are not scattered but simply not linear in ε. (b) The same barriers against $f(\varepsilon)$: solid, the fit to the five solvents; dashed, the fit including the gas phase. (c) TS bond distances and (d) reaction free energy against $f(\varepsilon)$; the vertical line in (d) marks where the reaction becomes exergonic.

It is worth being explicit about what this does and does not show. It does not show that $f(\varepsilon)$ is a better fitting function than ε in some statistical sense; with six points and two parameters that would mean little. It shows that the *same* six points give $R^2 \approx 0.5$ on one axis and $R^2 \approx 0.99$ on another, and that the second axis is the one the theory of the method says should appear. The statistics are a consequence of the physics, not a substitute for it.

3.2 A predictive model and its range of validity

The five solvents define

$$\Delta G^{\ddagger} = -32.2\, f(\varepsilon) + 43.6 \quad \text{kcal/mol}, \qquad R^2 = 0.996. \tag{6}$$

This is more than a description. It predicts the barrier in any solvent from its dielectric constant alone, without a further calculation. Table 3 tests it by leave-one-out cross-validation: each medium is predicted from a fit to the other five.

Table 3: Leave-one-out validation of Equation (6). Each row is predicted from a straight line fitted to the other five media.

Medium	$f(\varepsilon)$	predicted	calculated	error
			kcal/mol	
Gas	0.0000	43.59	47.67	−4.08
Heptane	0.1890	39.17	37.78	1.39
Benzene	0.2293	37.57	36.11	1.46
Chloroform	0.3561	32.53	31.68	0.85
Acetonitrile	0.4793	27.20	28.24	−1.04
Water	0.4905	26.62	27.97	−1.35

The five solvents are reproduced to within 1.5 kcal/mol. The gas phase is not: it is predicted 4.1 kcal/mol too low, and including it in the fit degrades R^2 from 0.996 to 0.977 and changes the slope from −32.2 to −38.9.

The deviation is not noise and it has a specific cause. Equation (3) is linear in $f(\varepsilon)$ *at constant* μ; the dipole moment of the TS is not constant, falling from 13.51 D in the gas phase to 11.96 D in water (Table 2). The gas phase is the point whose dipole departs furthest from the rest, and it deviates furthest from the line. The practical consequence is that Equation (6) should be used to interpolate between solvents and not to extrapolate to vacuum — which is also the regime in which a continuum model has nothing to describe.

3.3 The reactant complex is screened out of existence

The exercise assumes a reactant complex $H_3N{\cdots}CH_3Cl$ from which the barrier is measured. In the gas phase it exists, but barely: the complexation energy is −1.56 kcal/mol, consistent with a dipole–dipole interaction between two small neutral molecules. A dielectric medium screens such an interaction, and the screening is visible immediately (Table 4).

Table 4: Complexation energy and free energy of $H_3N{\cdots}CH_3Cl$ in the media where the complex is a minimum. Beyond benzene no minimum was located.

Medium	$f(\varepsilon)$	ΔE_{cplx}	ΔG_{cplx}
		kcal/mol	
Gas	0.0000	−1.56	3.24
Heptane	0.1890	−0.88	3.99
Benzene	0.2293	−0.73	4.21

In chloroform and above, the optimization does not converge, and the reason is instructive rather than technical. Over the last hundred optimization cycles the energy varies by less than 0.001 kcal/mol while the N–C distance drifts past 7 Å: the surface has become flat, and there is no minimum to converge to. The failure is a measurement of the screening, not an artefact of it. The same phenomenon was observed in session 4 for the ion–dipole complexes of $F^- + CH_3Cl$, which disappear in water; there the interaction was ion–dipole and survived to $\varepsilon = 78$, here it is dipole–dipole and does not survive $\varepsilon = 4.7$.

A second observation makes the choice of reference unavoidable. Even where the complex exists, its complexation *free* energy is positive: 3.2 kcal/mol to 4.2 kcal/mol. Associating two molecules into one costs translational and rotational entropy, about 6 kcal/mol at room temperature, which exceeds a binding energy of one or two. The complex is therefore above the separated reactants in free energy even when it is below them in electronic energy, and the separated reactants are the better reference on thermodynamic as well as practical grounds.

3.4 Geometry, charge and dipole: an earlier and less polarised TS

Two of the computed quantities behave opposite to naive expectation. As the medium becomes more polar, the charge on chlorine becomes *less* negative, from -0.790 to $-0.748\,e$, and the dipole moment of the TS *falls*, from 13.51 D to 11.96 D (Figure 3). The TS is less polarised in water than in vacuum.

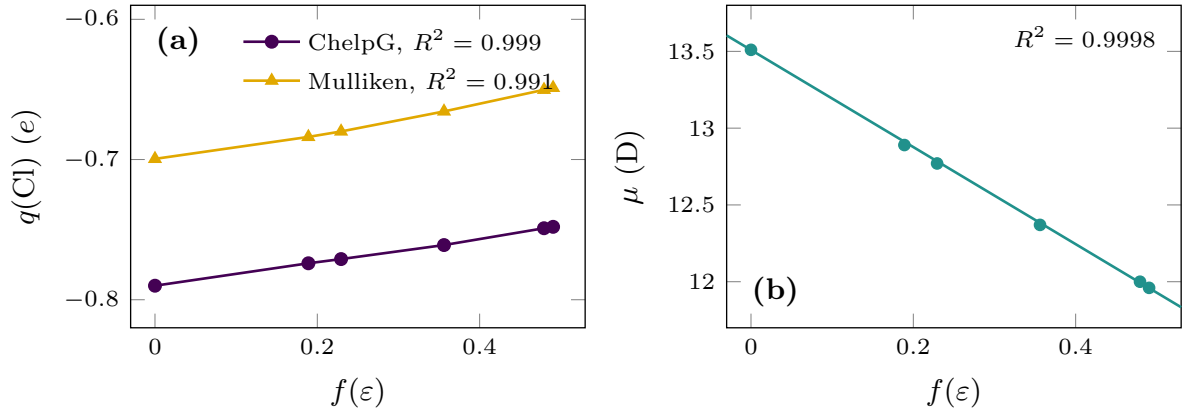


Figure 3: (a) Charge on chlorine against $f(\varepsilon)$ computed with ChelpG and with Mulliken population analysis. The absolute values differ by about $0.09\,e$, which is the arbitrariness of the convention, but the trend and the linearity against $f(\varepsilon)$ are the same. (b) Dipole moment of the TS, the most nearly linear of all the quantities computed.

A reaction field polarises a solute, so the direct effect of increasing ε should be to increase μ and the charge separation. Something larger is working in the opposite direction, and Table 2 identifies it: the geometry. The N–C distance lengthens from 1.899 Å to 2.209 Å while the C–Cl distance shortens from 2.474 Å to 2.289 Å. The TS moves *earlier* along the reaction coordinate as the solvent becomes more polar, and an earlier TS has transferred less charge. The geometric displacement overrides the direct polarisation.

That the dipole moment is the quantity that correlates best of all ($R^2 = 0.9998$) is worth noting, because it is also the quantity that appears explicitly in Equation (3). The chain $f(\varepsilon) \to$ reaction field $\to$ geometry $\to$ dipole closes on itself.

3.5 Hammond's postulate, measured

The reason the TS moves is thermodynamic. Figure 2d shows the reaction free energy: 114.09 kcal/mol in the gas phase, where the products are two separated ions in vacuum, falling to -14.93 kcal/mol in water. The solvent swings the thermodynamics of the reaction by 129 kcal/mol and turns it from impossible to spontaneous.

Hammond's postulate then requires the TS of a strongly endergonic reaction to resemble the products, and that of a less endergonic one to move back towards the reactants. Figure 4 tests this directly rather than by inspection.

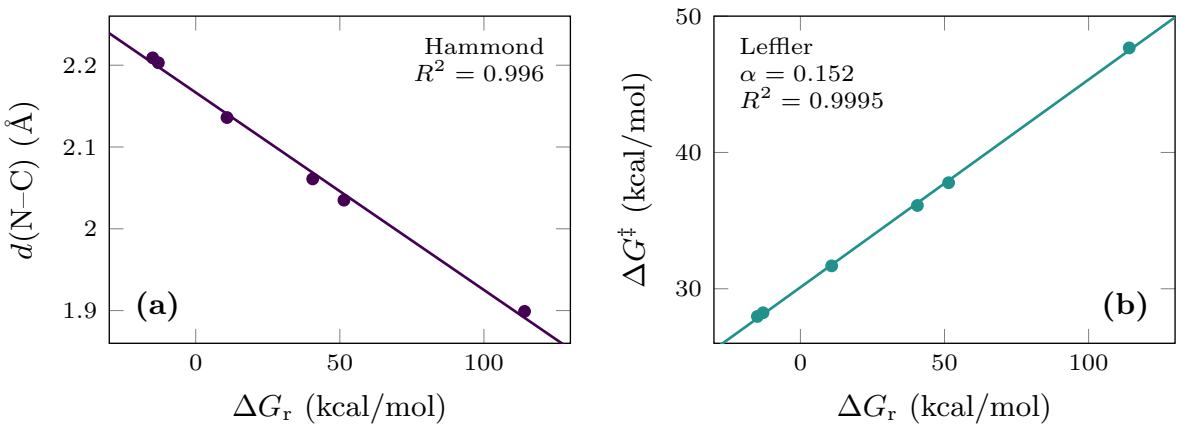


Figure 4: (a) Position of the transition state, measured by the forming N–C bond, against the reaction free energy: Hammond's postulate as a quantitative relation. (b) Barrier against reaction free energy; the slope is the Leffler coefficient α.

The forming bond tracks the thermodynamics with $R^2 = 0.996$, and the barrier tracks it with $R^2 = 0.9995$ and slope

$$\alpha = \frac{\partial \Delta G^‡}{\partial \Delta G_r} = 0.152, \tag{7}$$

the Leffler coefficient. A value near zero describes a reactant-like TS, one near unity a product-like one, and 0.15 places this TS early over the range sampled, consistent with the geometries.

Extrapolating the fit of Figure 2d, the reaction becomes exergonic at $f(\varepsilon) = 0.414$, that is $\varepsilon \approx 8$. This is a concrete and testable prediction, and it agrees with synthetic practice: Menshutkin alkylations are run in acetonitrile, DMSO or nitrobenzene, and not in hexane or toluene.

3.6 One variable, five correlations

The correlations above are not independent, and saying so is part of reporting them honestly.

In a continuum model the only property of the solvent that enters the calculation is ε, and it enters only through $f(\varepsilon)$. Every converged property of the solute — its geometry, its charge distribution, its energy — is therefore a function of that single variable. If each of them is linear in $f(\varepsilon)$ over the range sampled, then each is necessarily linear in each of the others. The Hammond relation of Figure 4a and the Leffler relation of Figure 4b follow from that common dependence; they do not constitute independent evidence for it.

What does carry physical content is not the existence of the correlations but their slopes. That $\alpha = 0.15$ rather than 0.85 is a statement about this transition state that no amount of common dependence could have produced; that the barrier falls by 32 kcal/mol across the full range of $f(\varepsilon)$ is a statement about this reaction.

The conclusion is also independent of the charge partition scheme, which is the obvious objection to any discussion built on atomic charges. Mulliken populations differ from ChelpG by about 0.09 e in absolute value, as expected from two different conventions, but give the same trend, the same sign and the same qualitative contrast between the two axes: $R^2 = 0.62$ against ε and 0.99 against $f(\varepsilon)$ (Figure 3a).

3.7 Comparison with experiment

The calculation predicts that the barrier falls by 9.5 kcal/mol between heptane and acetonitrile. Published Menshutkin rate data, converted to barrier differences with Equation (5), give about 5.6 kcal/mol over the same range of polarity (Table 5).

Table 5: Experimental relative rates for Menshutkin reactions [3] converted into barrier differences with Equation (5) at 298 K.

Solvent (relative to hexane)	ε	k_{rel}	$-\Delta\Delta G^{\ddagger}$ (kcal/mol)
Carbon tetrachloride	2.2	31	2.03
Chlorobenzene	5.6	1200	4.20
Acetonitrile	37.5	12 000	5.56
Dimethyl sulfoxide	46.7	50 000	6.41

The direction is right and the order of magnitude is right; the effect is overestimated by nearly a factor of two. Three contributions are expected, and it is worth separating them because they are not equally fundamental.

The first is the system. The experimental data are for tertiary amines and alkyl iodides, not for NH_3 and CH_3Cl; a more nucleophilic amine and a better leaving group give a different, and generally earlier, TS. This is a comparison of trends between related systems, not of absolute barriers, and only the trends should be expected to agree.

The second is the method. Hartree–Fock has no electron correlation, and there is no reason for the correlation energy to cancel between a pair of neutral closed-shell molecules and a strongly polarised TS. Correlation typically stabilises the more polarisable species, which here is the TS, so a correlated method would be expected to lower the barriers — but not uniformly across media, so the slope would change too.

The third is the solvent model, and it is the most interesting. A continuum contains no solvent molecules and therefore no specific hydrogen bonds. In protic media the developing chloride is stabilised by hydrogen bonding that PCM represents only as bulk polarisation of a dielectric. The effect of adding explicit solvent molecules would not be a uniform shift: it would act mainly at the polar end of the series, where it would stabilise the TS further.

3.8 What this analysis cannot say

Three limitations should be stated plainly.

The analysis is internal to the continuum approximation. It establishes that, within a model whose only solvent parameter is ε, the correct variable is $f(\varepsilon)$. It does not establish that real solvent effects are linear in $f(\varepsilon)$, and for protic solvents they are known not to be.

The cavity is not a sphere. Equation (3) was derived for a point dipole in a spherical cavity, whereas PCM uses a molecule-shaped cavity and the full charge distribution. That the numerical model reproduces the functional form of the analytical one is an empirical finding of this work, not a theorem; it holds because the boundary conditions are the same and the solute is reasonably compact.

Six points is not many. The conclusion rests on the contrast between $R^2 \approx 0.5$ and $R^2 \approx 0.99$ on the same data, which is large enough to be safe, but the slopes of the individual fits carry a correspondingly wide uncertainty that has not been propagated.

4 Conclusions

1. Correlating against ε, as the exercise requests, gives no linear relationship for any of the six quantities computed ($0.45 \le R^2 \le 0.58$). Correlating the same data against $f(\varepsilon) = (\varepsilon - 1)/(2\varepsilon + 1)$, the function in which ε actually appears in the theory of the method, gives a straight line for all of them ($0.977 \le R^2 \le 0.9998$). The apparent saturation of solvent effects is a property of the axis, not of the chemistry.

2. The resulting line, $\Delta G^{\ddagger} = -32.2\, f(\varepsilon) + 43.6$ kcal/mol, predicts the barrier in a solvent omitted from the fit to within 1.5 kcal/mol. It should not be extrapolated to the gas

phase, where the assumption of constant μ behind the linearity fails.

3. Since $f(\varepsilon)$ is the only solvent variable a continuum model contains, the several correlations requested by the exercise are in fact one correlation, and only their slopes carry independent information.

4. Those slopes are chemically sensible. The solvent swings the reaction free energy by 129 kcal/mol, turning the reaction exergonic above $\varepsilon \approx 8$, and displaces the transition state earlier along the reaction coordinate (Leffler $\alpha = 0.15$), which leaves it *less* polarised in water than in vacuum — the opposite of what direct polarisation of the solute alone would give.

5. The reactant complex assumed by the exercise is screened out of existence above $\varepsilon \approx 2.3$, and is above the separated reactants in free energy even where it exists. Barriers must be referred to the separated reactants, which is also what the second-order experimental kinetics require.

6. Against experiment the model reproduces the direction and the order of magnitude of the solvent effect (9.5 against 5.6 kcal/mol between heptane and acetonitrile) but overestimates it, as expected from a method without electron correlation and a continuum model without hydrogen bonds.

A Calculations that failed, and what they showed

Two of the 35 jobs attempted did not converge. Both are reported because in one case the failure was the result.

The reactant complex beyond benzene. The chained optimization terminated in link 9999 after exhausting 100 cycles. Inspection of the trajectory showed the energy constant to within 0.001 kcal/mol over the last hundred steps while d(N–C) drifted past 7 Å. No tighter convergence criterion or larger cycle limit would have helped: there was no stationary point. This is reported in Section 3.3 as a result.

The methylammonium cation beyond heptane. The same link-9999 termination after 38 cycles, but here the cause was numerical. The job had been set up with a plain `opt` keyword and an estimated Hessian, which is adequate in the gas phase but not on the flatter surface produced by a reaction field. Repeating the four remaining media with `opt=(calcfc,maxcycles=200)`, so that the Hessian is computed rather than guessed, converged all of them without further difficulty.

The contrast between the two is the practical lesson: an optimization that fails because the surface is flat and one that fails because the optimizer is badly initialised produce the same error message, and are distinguished by looking at the energy along the trajectory rather than at the final geometry.

B Extraction script

The following `awk` program reads any number of Gaussian output files and emits one record per converged job. It is reproduced here because the choice of patterns encodes several non-obvious features of the output format.

```
/^ Solvent  / {
    sol = $3; sub(/,$/, "", sol)
    split($0, p, "Eps="); eps = p[2] + 0
}
/^ SCF Done:/            { e = $5 }
```

```
$1 == "!" && $2 == "r1"  { rnc  = $3 }
$1 == "!" && $2 == "r5"  { rccl = $3 }
/^ ESP charges:$/        { inesp = 1; next }
inesp && $2 == "Cl"      { q = $3; inesp = 0 }
/Dipole moment \(field-independent/ { getline; mu = $NF }
/^ Frequencies --/       { if (nu == "") nu = $3 }
/Sum of electronic and thermal Free Energies=/ {
    print ...; sol = ""; eps = 0; rnc = ""; rccl = ""; q = ""; nu = ""
}
```

Three points are worth noting. The solvent and its dielectric constant are read from the `Solvent` banner rather than from the route line, so that the value used is the one the program actually applied. The pattern for the charges is anchored at both ends, `/^ ESP charges:$/`, because Gaussian prints three blocks whose headers all contain that string and only one of which is wanted. And the record is emitted on the thermochemistry line rather than on `Normal termination`, because a job that failed never reaches the former — which is why the failed jobs simply do not appear in the output rather than appearing with corrupt values.

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